

SOHAIL HARESH GIDWANI

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EXPERIENCE

Keck School of Medicine, USC

Oct 2025 - Present

Student Worker / Research Assistant, Multi-Modal AI for Alzheimer's Disease

Los Angeles, CA

- Architected a multi-modal deep learning pipeline (T1 MRI, DTI, clinical) across 2,363 ADNI subjects with cross-attention fusion and modality dropout, so one set of weights runs on any subset of inputs; achieved 68.2% balanced accuracy on 3-class diagnosis and 91.3% on binary classification (CN vs Dementia), and 78.7% zero-shot on the external OASIS-3 cohort with DTI entirely absent.
- Built a VQA pipeline on the frozen VLM (FAISS retrieval, cross-encoder reranking, Mistral 7B / Gemma 4 26B MoE / MedGemma 1.5 4B); a leakage audit traced its headline accuracy to diagnosis labels in retrieved captions, so the encoder diagnoses and the LLM layer explains.
- Owned end-to-end experimentation infrastructure in PyTorch: two-stage training (CLIP contrastive pre-training → multi-task fine-tuning), 7-way modality ablations with paired-bootstrap significance tests, and confidence calibration; first-author paper accepted at *Frontiers in Computational Neuroscience* (2026). [link]

Insaito, Inc.

May 2025 - Jul 2025

Senior Software Engineer - I

Remote

- Architected an AI agent platform in TypeScript and Python on serverless cloud infrastructure using LangChain and LangGraph for stateful, graph-based multi-step agent workflows across 100+ third-party integrations, with task routing, retries, and persistent context across long-running runs.
- Built event-driven backend services for concurrent agent orchestration with tool-calling, structured fallbacks, and a production tracing layer over agent outputs; owned core infrastructure for open-source LLM deployment, MCP servers, and prompt/evaluation harnesses for long tool-use chains.

IIFL Finance Ltd.

Jun 2023 - May 2025

Full Stack Software Developer

Mumbai, India

- Shipped production AI systems across two years at India's largest non-banking financial company, working on agent-driven customer interaction and workflow automation in a regulated financial services environment with integration into legacy core systems; embedded with the teams who would use each system before building.
- Designed CapitalGenie, a GPT-4o-powered support agent with custom prompt engineering and internal API orchestration that ingested unstructured user issue data, structured it, and generated personalized responses, accelerating resolution time by 70%.
- Built an internal RAG chatbot with a React frontend and a Python/Flask backend from scratch (Qdrant, Azure OpenAI), integrated into the Zoho ticketing flow and cutting support tickets by 20–25% in the first month. (Certificate of Achievement)
- Engineered an AI-powered Gold Loan Image Audit pipeline using GroundingDINO and Swin-Transformer for automated fraud detection on loan application photos, reducing potential loan fraud by 15%.

EDUCATION

University of Southern California

Aug 2025 - May 2027

M.S., Computer Science, GPA: 3.75/4.0

Los Angeles, CA

- **Coursework:** Applied NLP, Analysis of Algorithms, Information Retrieval & Web Search Engines, ML for Data Science

University of Mumbai

Aug 2019 - May 2023

B.E., Computer Engineering, CGPA: 9.05/10

Mumbai, India

SKILLS

AI/ML: PyTorch, TensorFlow, HuggingFace, scikit-learn, pandas, numpy, LangChain, LangGraph, Claude Code, fine-tuning, RAG, FAISS, cross-encoder reranking, agentic workflows, MCP, prompt engineering, LLM evaluation & benchmarking (Llama 3.2 3B, Qwen 2.5 7B, Mistral 7B, Gemma 4 26B MoE, MedGemma 1.5 4B), Vector DBs

Programming: Python, TypeScript, JavaScript, Java, C/C++, SQL

Backend & Cloud: FastAPI, Flask, Node.js, Express, Hono, RESTful APIs, AWS, Azure, GCP, Docker, Linux, Serverless, event-driven architectures

Databases & Tooling: PostgreSQL, pgvector, MySQL, MongoDB, Oracle, Qdrant, SQLAlchemy, Git, CI/CD, Test-Driven Development, Code Reviews

PROJECTS

- **Portage: Autonomous Flask→FastAPI Migration Agent** (*in active development*): LangGraph agent that plans and rewrites multi-file migrations, designs the new modules a migration requires, judges itself by running each repo's own test suite in a network-off Docker sandbox, and recovers from failures within bounded budgets; measured at K-run mean and variance, two apps that had never gone green now pass 5/5 autonomous runs each at K=5 (24/24 and 15/15 tests). [GitHub]
- **CoT Faithfulness Analysis (USC NLP):** Designed a 4-experiment study of whether chain-of-thought causally drives LLM answers or is post-hoc rationalization. Ran ~15K queries against Llama 3.2 3B and Qwen 2.5 7B on GSM8K and ARC-Challenge; found math CoT partially faithful (44–50pp drop under corruption) while science multiple-choice CoT is largely decorative. [link]
- **Knowledge Hub (RAG-Powered Q&A Agent):** Dockerized RAG agent on Flask and PostgreSQL+pgvector with hybrid full-text + vector retrieval, a citation-enforced Q&A pipeline via a local LLM, and multi-pass OCR for PDFs, images, and handwritten notes. [link]